Advanced Notification Routing

> **Series:** WFLOW | **Notebook:** 4 of 9 | **Created:** January 2026 | **Last Updated:** 01/28/2026

Intelligent Alert Routing

Not all alerts should go to everyone. This notebook covers conditional routing based on severity, team ownership, time of day, and escalation patterns.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Platform subscription
Permissions	`automation:workflows:write`
Prior Knowledge	**WFLOW-01** through **WFLOW-03**
Connections	Slack and/or Teams connections configured

1. Routing Strategies

Why Route Alerts?

Problem	Impact	Solution
Alert fatigue	Teams ignore alerts	Route only relevant alerts
Slow response	Wrong team notified	Route to owners
Off-hours noise	Sleep disruption	Time-based routing
Missed escalation	Unacknowledged alerts	Auto-escalation

Routing Dimensions

Dimension	Route Based On	Example
Severity	Problem severity level	Critical → PagerDuty, Low → Slack
Team	Entity tags, management zones	`team:checkout` → #checkout-alerts
Service	Service name or entity ID	Payment service → payments team
Time	Hour of day, day of week	Weekends → on-call only
Environment	prod/staging/dev tags	Prod → immediate, Dev → daily digest

2. Conditional Expressions

Conditions control which tasks execute. They use Jinja expressions returning boolean values.

Defining Conditions

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'

  - name: is_production
    expression: '{{ "prod" in event()["management_zones"] }}'

  - name: is_checkout_team
    expression: '{{ "team:checkout" in event().get("tags", []) }}'
```

Using Conditions on Tasks

```
tasks:
  - name: pagerduty_alert
    type: dynatrace.pagerduty:create-incident
    conditions:
      - is_critical
      - is_production
    # Task runs only if BOTH conditions are true (AND logic)
```

Negating Conditions

```
tasks:
  - name: slack_only_for_non_critical
    type: dynatrace.slack:message
    conditions:
      - not is_critical
    # Task runs when is_critical is FALSE
```

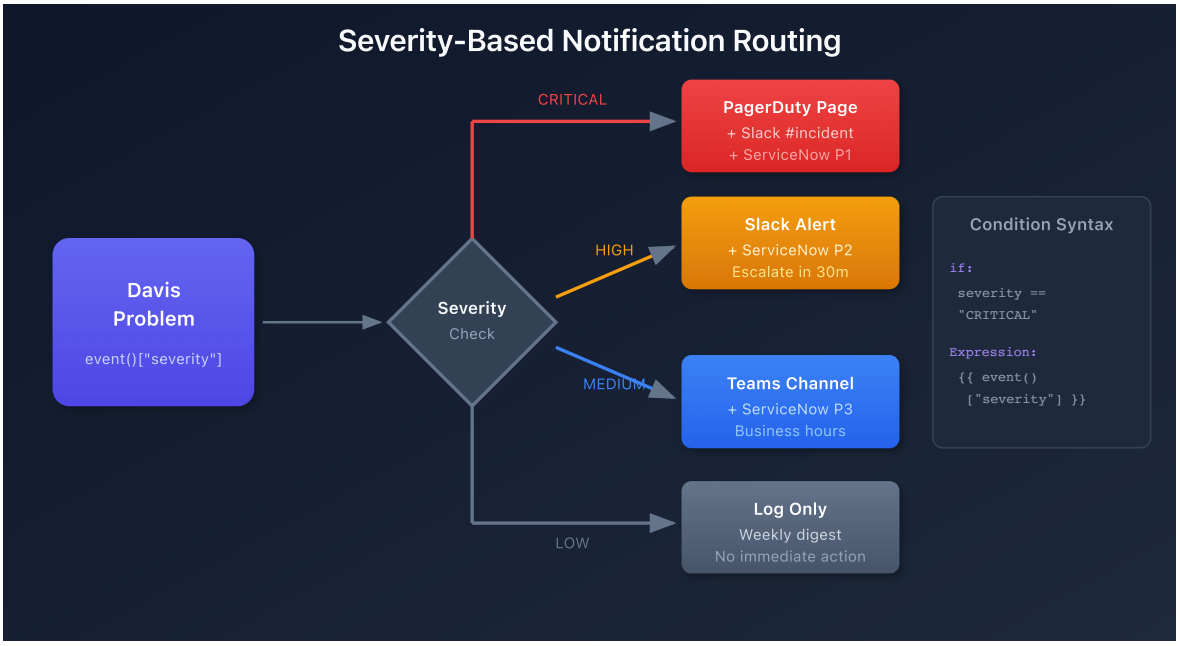
Complex Expressions

```
# AND within expression
{{ event()["severity"] == "CRITICAL" and "prod" in event()
["management_zones"] }}

# OR within expression
{{ event()["severity"] in ["CRITICAL", "HIGH"] }}

# Check if field exists
{{ event().get("root_cause_entity_id") is not none }}
```

3. Routing by Severity

Severity-Based Routing Pattern

Severity	Actions
CRITICAL	PagerDuty + Slack #urgent + Email
HIGH	Slack #alerts + Email
MEDIUM	Slack #alerts
LOW	Slack #alerts (business hours only)



Workflow Configuration

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'
  - name: is_high
    expression: '{{ event()["severity"] == "HIGH" }}'
  - name: is_medium_or_low
    expression: '{{ event()["severity"] in ["MEDIUM", "LOW"] }}'

tasks:
  # CRITICAL: Page on-call + Slack urgent + Email
  - name: pagerduty_critical
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical]
    input:
      connection: pagerduty-prod
      severity: critical
      summary: "{{ event()['title'] }}"

  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical]
    input:
      connection: slack-production
      channel: "#alerts-urgent"
      message: ":rotating_light: *CRITICAL* {{ event()['title'] }}"

  # HIGH: Slack alerts + Email
  - name: slack_high
    type: dynatrace.slack:message
    conditions: [is_high]
    input:
      channel: "#alerts-production"
      message: ":warning: *HIGH* {{ event()['title'] }}"

  # MEDIUM/LOW: Slack only
  - name: slack_low
    type: dynatrace.slack:message
    conditions: [is_medium_or_low]
    input:
      channel: "#alerts-production"
      message: ":information_source: *{{ event()['severity'] }}* {{ event()
['title'] }}"
```

4. Routing by Team/Service

Tag-Based Team Routing

Entities tagged with `team:checkout`, `team:payments`, etc.

```
conditions:
  - name: is_checkout_team
    expression: |
      {% set tags = event().get("tags", []) %}
      {{ "team:checkout" in tags or "team:cart" in tags }}

  - name: is_payments_team
    expression: '{{ "team:payments" in event().get("tags", []) }}'

  - name: is_platform_team
    expression: '{{ "team:platform" in event().get("tags", []) }}'

tasks:
  - name: notify_checkout
    type: dynatrace.slack:message
    conditions: [is_checkout_team]
    input:
      channel: "#checkout-alerts"
      message: "{{ event()['title'] }}"

  - name: notify_payments
    type: dynatrace.slack:message
    conditions: [is_payments_team]
    input:
      channel: "#payments-alerts"
      message: "{{ event()['title'] }}"

  - name: notify_platform
    type: dynatrace.slack:message
    conditions: [is_platform_team]
    input:
      channel: "#platform-alerts"
      message: "{{ event()['title'] }}"
```

Management Zone Routing

```
conditions:
  - name: is_us_region
    expression: '{{ "US-East" in event()["management_zones"] or "US-West" in
event()["management_zones"] }}'

  - name: is_eu_region
    expression: '{{ "EU-West" in event()["management_zones"] }}'

tasks:
  - name: notify_us_team
    conditions: [is_us_region]
    input:
      channel: "#us-oncall"

  - name: notify_eu_team
    conditions: [is_eu_region]
    input:
      channel: "#eu-oncall"
```

Dynamic Channel Selection

Use JavaScript to determine the channel dynamically:

```
export default async function({ event }) {
  const tags = event.tags || [];

  // Find team tag
  const teamTag = tags.find(t =&gt; t.startsWith('team:'));
  const team = teamTag ? teamTag.split(':')[1] : 'platform';

  return {
    channel: `#${team}-alerts`,
    team: team
  };
}
```

5. Time-Based Routing

Business Hours vs Off-Hours

```
conditions:
  - name: is_business_hours
    expression: |
      {% set hour = now().hour %}
      {% set weekday = now().weekday() %}
      {{ weekday &lt; 5 and hour &gt;= 9 and hour &lt; 17 }}

  - name: is_off_hours
    expression: |
      {% set hour = now().hour %}
      {% set weekday = now().weekday() %}
      {{ weekday &gt;= 5 or hour &lt; 9 or hour &gt;= 17 }}

tasks:
  # Business hours: Slack channel
  - name: slack_business_hours
    type: dynatrace.slack:message
    conditions: [is_business_hours]
    input:
      channel: "#alerts-production"
      message: "{{ event()['title'] }}"

  # Off-hours: PagerDuty for CRITICAL only
  - name: pagerduty_off_hours
    type: dynatrace.pagerduty:create-incident
    conditions: [is_off_hours, is_critical]
    input:
      connection: pagerduty-prod
      summary: "[Off-Hours] {{ event()['title'] }}"

  # Off-hours non-critical: Queue for morning
  - name: slack_queue
    type: dynatrace.slack:message
    conditions: [is_off_hours, not is_critical]
    input:
      channel: "#alerts-queue"
      message: ":moon: *Queued for review* {{ event()['title'] }}"
```

Weekend-Only Routing

```
conditions:
  - name: is_weekend
    expression: '{{ now().weekday() &gt;= 5 }}'

tasks:
  - name: weekend_oncall
    conditions: [is_weekend, is_critical]
    input:
      # Page weekend on-call rotation
      routing_key: "{{ env.PD_WEEKEND_ROUTING_KEY }}"
```

6. Escalation Patterns

Timed Escalation

Escalate if not acknowledged within a time window.

```
tasks:
  # Step 1: Initial notification
  - name: initial_slack
    type: dynatrace.slack:message
    input:
      channel: "#alerts-production"
      message: ":warning: {{ event()['title'] }} - Acknowledge within 15 min"

  # Step 2: Wait 15 minutes
  - name: wait_for_ack
    type: dynatrace.automations:wait
    dependsOn: [initial_slack]
    input:
      duration: "15m"

  # Step 3: Check if still open
  - name: check_problem_status
    type: dynatrace.automations:run-javascript
    dependsOn: [wait_for_ack]
    input:
      script: |
        import { eventsClient } from '@dynatrace-sdk/client-classic-
environment-v2';

        export default async function({ event }) {
          const problemId = event.display_id;
          // Check if problem is still open
          // Return escalate: true if needs escalation
          return { escalate: event.status === 'OPEN' };
        }

  # Step 4: Escalate to PagerDuty
  - name: escalate_pagerduty
    type: dynatrace.pagerduty:create-incident
    dependsOn: [check_problem_status]
    conditions:
      - '{{ result("check_problem_status").escalate }}'
    input:
      severity: high
      summary: "[ESCALATED] {{ event()['title'] }} - No acknowledgment in 15
min"
```

Multi-Tier Escalation

```
0 min  → Slack channel notification
15 min → Email to team lead
30 min → PagerDuty to on-call
60 min → PagerDuty to manager
```

7. Multi-Channel Strategy

Recommended Channel Matrix

Severity	Environment	Channel(s)
CRITICAL	Production	PagerDuty + Slack urgent + Email
CRITICAL	Staging	Slack alerts
HIGH	Production	Slack alerts + Email
HIGH	Staging	Slack alerts
MEDIUM	Production	Slack alerts
MEDIUM	Staging	Slack (daily digest)
LOW	Any	Slack (weekly digest)

Complete Multi-Channel Workflow

```
conditions:
  - name: is_critical
    expression: '{{ event()["severity"] == "CRITICAL" }}'
  - name: is_high_or_above
    expression: '{{ event()["severity"] in ["CRITICAL", "HIGH"] }}'
  - name: is_production
    expression: '{{ "Production" in event().get("management_zones", []) }}'
  - name: is_business_hours
    expression: '{{ now().weekday() &lt; 5 and now().hour &gt;= 9 and
now().hour &lt; 17 }}'

tasks:
  # PagerDuty: Critical + Production
  - name: pagerduty
    type: dynatrace.pagerduty:create-incident
    conditions: [is_critical, is_production]

  # Slack Urgent: Critical + Production
  - name: slack_urgent
    type: dynatrace.slack:message
    conditions: [is_critical, is_production]
    input:
      channel: "#alerts-urgent"

  # Slack Standard: High or above + Production
  - name: slack_standard
    type: dynatrace.slack:message
    conditions: [is_high_or_above, is_production]
    input:
      channel: "#alerts-production"

  # Email: Critical or (High + Production + Business Hours)
  - name: email_alert
    type: dynatrace.email:send
    conditions:
      - '{{ event()["severity"] == "CRITICAL" or (event()["severity"] ==
"HIGH" and "Production" in event().get("management_zones", [])) }}'
```

Query Workflow Routing Effectiveness

```
// Workflow executions by trigger severity
fetch events, from: now() - 7d
| filter event.type == "automation.workflow.execution"
| summarize executions = count(), by:{workflow.name, trigger.severity}
| sort executions desc
| limit 30
```

```
// Task execution distribution by notification channel
fetch events, from: now() - 7d
| filter event.type == "automation.task.execution"
| filter contains(task.type, "slack") or contains(task.type, "msteams") or
contains(task.type, "pagerduty") or contains(task.type, "email")
| summarize
    total = count(),
    succeeded = countIf(task.status == "SUCCEEDED"),
    by:{task.type}
| fieldsAdd success_rate = round(100.0 * succeeded / total, decimals: 2)
| sort total desc
```

```
// Skipped tasks (conditions not met)
fetch events, from: now() - 24h
| filter event.type == "automation.task.execution"
| filter task.status == "SKIPPED"
| summarize skipped_count = count(), by:{workflow.name, task.name}
| sort skipped_count desc
| limit 20
```

Next Steps

With routing configured, integrate with incident management:

Recommended Path

1. **WFLOW-05: PagerDuty & ServiceNow** - Create incidents automatically
2. **WFLOW-06: Custom Templates** - Rich message formatting
3. **WFLOW-07: Problem-Triggered Remediation** - Auto-remediation

Key Takeaways

- **Conditions** control task execution using Jinja expressions
- **Severity routing** ensures appropriate response levels
- **Team routing** uses tags or management zones
- **Time-based routing** reduces off-hours noise
- **Escalation patterns** prevent missed alerts
- Test conditions with On-Demand trigger before production

Summary

In this notebook, you learned:

- Why and when to route alerts conditionally
- Conditional expression syntax and operators
- Severity-based routing patterns
- Team and management zone routing
- Time-based (business hours) routing
- Escalation with wait and check patterns
- Multi-channel notification strategy

References

- Workflow Conditions
- Jinja Expressions
- Wait Action
- Task Dependencies

Dynatrace documentation.